Name
Date
Course

Complex Fractions (Homework)

Simplify the following complex fractions. Answers

1. $$\frac{\frac{2}{7}}{\frac{3}{4}}$$ $\boxed{\frac{8}{21}}$

2. $$\frac{\frac{1}{5}}{\frac{7}{15}}$$ $\boxed{\frac{3}{7}}$

3. $$\frac{\frac{3}{4}}{-\frac{1}{2}}$$ $\boxed{-\frac{3}{2}}$

4. $$\frac{-\frac{5}{6}}{-\frac{1}{4}}$$ $\boxed{\frac{10}{3}}$

5. $$\frac{\frac{1}{6}}{\frac{5}{8x}}$$ $\boxed{\frac{4x}{15}}$

6. $$\frac{\frac{3}{x^2y}}{-\frac{4}{y^2}}$$ $\boxed{\frac{3y}{-4x^2}}$

7. $$\frac{\frac{8}{bc}}{\frac{4}{b}}$$ $\frac{8}{4c} = \boxed{\frac{2}{c}}$

8. $$\frac{\frac{-2}{5x^2}}{\frac{3}{xy}}$$ $\boxed{-\frac{2y}{15x}}$

9. $$\frac{\frac{8}{b}}{\frac{3}{b-5}}$$ $\boxed{\frac{8b-40}{3b}}$

10. $$\frac{\frac{-2}{y-3}}{\frac{7}{y^2-8y+15}}$$ $\boxed{\frac{-2y+10}{7}}$

(11) $$\frac{\frac{8x}{4+x}}{\frac{5}{x^2-16}}$$

$$\boxed{\frac{8x^2-32x}{5}}$$

(12) $$\frac{\frac{5}{27c^2+36c}}{\frac{2}{9c}}$$

$$\boxed{\frac{5}{6c+8}}$$

(13) $$\frac{\frac{5}{b^2+11b+24}}{\frac{-2}{b+8}}$$

$$-\frac{5}{2b+6}$$ or $$\frac{-5}{2b+6}$$ or $$\frac{5}{-2b-6}$$

(14) $$\frac{\frac{1}{3}+\frac{7}{15}}{\frac{8}{5}}$$

$$\boxed{\frac{1}{2}}$$

(15) $$\frac{\frac{3}{8}}{\frac{5}{8}+\frac{1}{4}}$$

$$\boxed{\frac{3}{7}}$$

(16) $$\frac{\frac{3}{4}-\frac{2}{7}}{\frac{6}{7}+\frac{1}{28}}$$

$$\boxed{\frac{13}{25}}$$

(17) $$\frac{\frac{11}{18}+\frac{7}{9}}{\frac{1}{6}-\frac{2}{9}}$$

$$\boxed{-25}$$

(18) $$\frac{\frac{2}{x}-\frac{3}{5x^2}}{\frac{1}{5}-\frac{1}{x^2}}$$

$$\boxed{\frac{10x-3}{x^2-5}}$$

(19) $$\frac{\frac{3}{8}+\frac{2}{5y^2}}{\frac{9}{20y}-\frac{7}{40}}$$

$$\boxed{\frac{15y^2+16}{18y-7y^2}}$$

20. $$\frac{\frac{7}{x-1}+\frac{2}{x+1}}{\frac{3x}{x^2-1}}$$ $\boxed{\dfrac{9x+5}{3x}}$

21. $$\frac{\frac{2}{x+6}+\frac{5x}{x^2+x-30}}{\frac{1}{x-5}}$$ $\boxed{\dfrac{7x-10}{x+6}}$

22. $$\frac{\frac{5}{3y^2-6y}+\frac{4}{3y}}{\frac{y}{3}-\frac{6}{y-2}}$$ $\boxed{\dfrac{4y-3}{y^3-2y^2-18y}}$

23. $$\frac{\frac{8}{b+5}-\frac{2}{b+1}}{8+\frac{3}{b^2+6b+5}}$$ $\boxed{\dfrac{6b-2}{8b^2+48b+43}}$

24. $$\frac{\frac{1}{x+6}+5}{\frac{3}{x+6}-1}$$ $\boxed{\dfrac{5x+31}{-x-3}}$

25. $$\frac{\frac{1}{y-3}+2}{y^2+\frac{4}{y+7}}$$ $\boxed{\dfrac{2y^2+9y-35}{y^4+4y^3-21y^2+4y-12}}$

26. $$\frac{\frac{2}{2b-3}-\frac{b}{b-4}}{\frac{10}{b-4}-\frac{9}{2b^2-11b+12}}$$ $\boxed{\dfrac{-2b^2+5b-8}{20b-39}}$

27. $$\frac{\frac{4}{5}}{2}$$ $\boxed{\dfrac{2}{5}}$

28. $$\frac{6}{\frac{1}{4}}$$ $\boxed{24}$

29. $$\frac{7x^2}{\frac{2}{3x}}$$ $\boxed{\dfrac{21x^3}{2}}$

30. $$\frac{\frac{5x}{6}}{2x}$$ $\boxed{\dfrac{5}{12}}$